

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

The dataset was analyzed and prepared using Python (**Pandas**) before performing SQL analysis and building the Power BI dashboard.

Data Inspection:

- Loaded the dataset into Pandas.
- Examined the first and last records using `head()` and `tail()`.
- Checked dataset dimensions, column names, and data types.
- Generated descriptive statistics to understand the distribution of variables.
- Inspected data types, non-null values, and memory usage using `info()`.

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RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Customer ID                          3900 non-null   int64
 1   Age                                  3900 non-null   int64
 2   Gender                              3900 non-null   object
 3   Item Purchased                      3900 non-null   object
 4   Category                            3900 non-null   object
 5   Purchase Amount (USD)               3900 non-null   int64
 6   Location                            3900 non-null   object
 7   Size                                3900 non-null   object
 8   Color                               3900 non-null   object
 9   Season                              3900 non-null   object
10   Review Rating                       3863 non-null   float64
11   Subscription Status                 3900 non-null   object
12   Shipping Type                      3900 non-null   object
13   Discount Applied                   3900 non-null   object
14   Promo Code Used                    3900 non-null   object
15   Previous Purchases                 3900 non-null   int64
16   Payment Method                    3900 non-null   object
17   Frequency of Purchases             3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB

```

- Generated descriptive statistics using `describe()` to understand the distribution of numerical and categorical variables.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Data Quality Checks:

- Identified missing values using `isnull()`.
- Checked for duplicate records using `duplicated()`.

- Found missing values only in the **Review Rating** column.
- No duplicate records were found.

Data Cleaning:

- Filled missing values in **Review Rating** using the **median rating of each product category**.
- Renamed column names to a consistent lowercase and underscore (**snake_case**) format.
- Renamed **purchase_amount_(usd)** to **purchase_amount** for better readability.
- Removed the **promo_code_used** column after confirming it contained the same information as **discount_applied**.

Feature Engineering:

- Created a new Age Group column by dividing customers into 'Young Adult', 'Adult', 'Middle-aged' and 'Senior'.
- Converted purchase frequency (e.g., Weekly, Monthly, Quarterly) into a numeric **purchase_frequency_days** column to simplify future analysis.

Database Integration:

- Connected the cleaned dataset to a MySQL database for SQL-based business analysis and querying.

4. SQL Business Analysis:

Revenue by Gender - Compared total revenue generated by male and female customers.

gender	total_revenue
Male	157890
Female	75191

High-Spending Customers Using Discounts - Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88

839 row(s) returned

Top 5 Products by Average Review Rating - Identified the highest-rated products based on customer reviews.

item_purchased	average_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

Purchase Amount by Shipping Type - Compared the average purchase amount between Standard and Express shipping.

shipping_type	average_purchase_amount
Express	60.48
Standard	58.46

Subscriber vs. Non-Subscriber Spending - Compared customer count, average spending, and total revenue based on subscription status.

subscription_status	total_customers	average_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

Products with Highest Discount Usage - Identified 5 products, most frequently purchased with discounts.

item_purchased	percentage_of_purchases
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Customer Segmentation - Classified customers as New, Returning, or Loyal based on previous purchases.

customer_segment	number_of_customers
loyal	3116
returning	701
new	83

Top 3 Products in Each Category - Determined the most purchased products within each product category.

category	item_purchased	total_orders	item_rank
Accessories	Jewelry	171	1
Accessories	Sunglasses	161	2
Accessories	Belt	161	3
Clothing	Blouse	171	1
Clothing	Pants	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

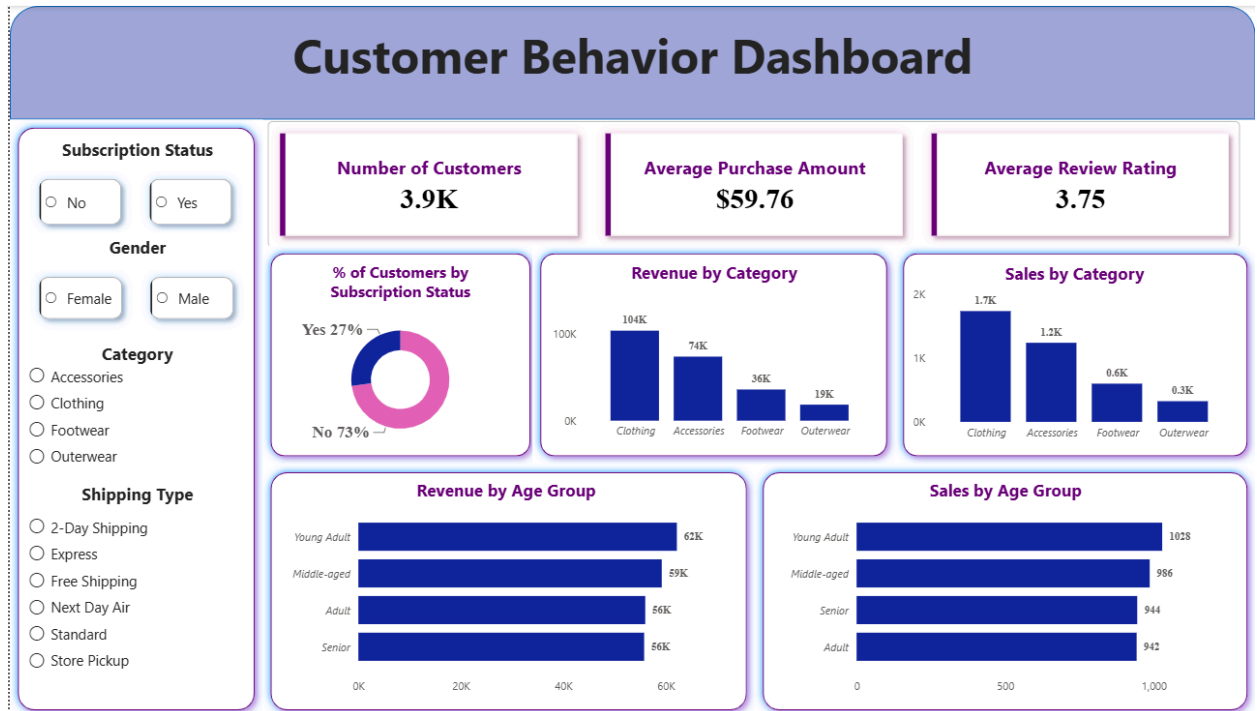
Repeat Buyers and Subscription Status - Analyzed whether repeat buyers are more likely to subscribe.

subscription_status	repeat_buyers
Yes	958
No	2518

Revenue Contribution by Age Group - Compared total revenue generated by different age groups.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Power BI Dashboard



6. Business Recommendations

Increase Customer Subscriptions - Promote exclusive discounts and loyalty benefits to encourage more customers to subscribe and improve long-term customer retention.

Strengthen Customer Loyalty - Reward repeat customers with personalized offers or loyalty programs to increase repeat purchases and customer lifetime value.

Optimize Discount Strategy - Review discount campaigns regularly to ensure they drive sales while maintaining healthy profit margins.

Promote Top-Performing Products - Highlight best-selling and highly rated products in marketing campaigns to maximize customer engagement and conversions.

Focus on High-Value Customer Segments - Target marketing efforts toward the highest-revenue age groups and customer segments to improve campaign effectiveness.